

★ WEEK 11 · PART 3 of 3

OU in Practice: Pairs Trading Stage V

A Principled Mean-Reversion Model of the Spread

OU model of the spread · estimation · half-life filter · the cautions

The OU process finally gives the pairs spread a model, not just thresholds: a fair value, statistically-grounded bands, and a half-life. We estimate it, trade it, and confront what can go wrong.

3

PART 3 · IN PRACTICE

An OU Model of the Spread

Pairs trading Stage V models the spread as an OU process, estimated through the AR(1) bridge, with the half-life as a trade filter — plus the cautions, and the road to interest-rate models.

From the Model to the Trade

Parts 1-2 built OU; Part 3 uses it

1

We have a mean-reversion model

The OU process, with a spring toward the mean, three parameters, and a half-life — and it discretizes to an AR(1).

2

The pairs spread reverts

A cointegrated spread is exactly the kind of mean-reverting series OU was made for. So model it as OU.

3

Stage V makes the trade principled

Fair value, entry bands, and a horizon all fall out of the fitted OU parameters — replacing ad-hoc thresholds with an estimable model.

Part 3 in one line: model the spread as OU, estimate it, and let its parameters set the trade.

Stage V: An OU Model of the Spread

The thread gets a principled mean-reversion model

Earlier stages built the spread and protected it. Stage V finally MODELS it. Treating the spread as an OU process gives a fair value, bands grounded in its statistics, and a half-life — all estimable from data.

What we had

- A validated, mean-reverting spread (Stage II).
- An adaptive hedge and a regime filter (III, IV).
- But entry rules were still z-score thresholds.

What Stage V adds

- A model: the spread IS an OU process.
- Fair value, bands, and horizon from its parameters.
- A principled, estimable basis for the trade.

Why Model the Spread as OU

From thresholds to a model

The problem with thresholds

- A z-score of two is an arbitrary line.
- It says nothing about how fast reversion happens.
- It gives no horizon and no fair value estimate.

What OU provides

- A fair value: the reversion level θ .
- Bands from the spread's own long-run spread.
- A half-life that sets the expected holding time.

The Spread as an OU Process

One SDE for the whole trade

model the spread as OU: $d(\text{spread}_t) = \kappa(\theta - \text{spread}_t) dt + \sigma dW_t$

Reading it

- The spread is pulled toward theta at speed kappa.
- Theta is the fair value — where it belongs.
- Sigma sets how much it wanders around theta.

Why it fits

- A cointegrated spread reverts, exactly as OU does.
- The tether from Week 7 becomes a spring here.
- The model is simple, and its parameters are estimable.

From OU to Trading Bands

Bands grounded in the spread's statistics

$$\text{bands: enter when } |\text{spread} - \theta| > z \sigma_{eq}, \quad \sigma_{eq} = \frac{\sigma}{\sqrt{2\kappa}}$$

The entry band

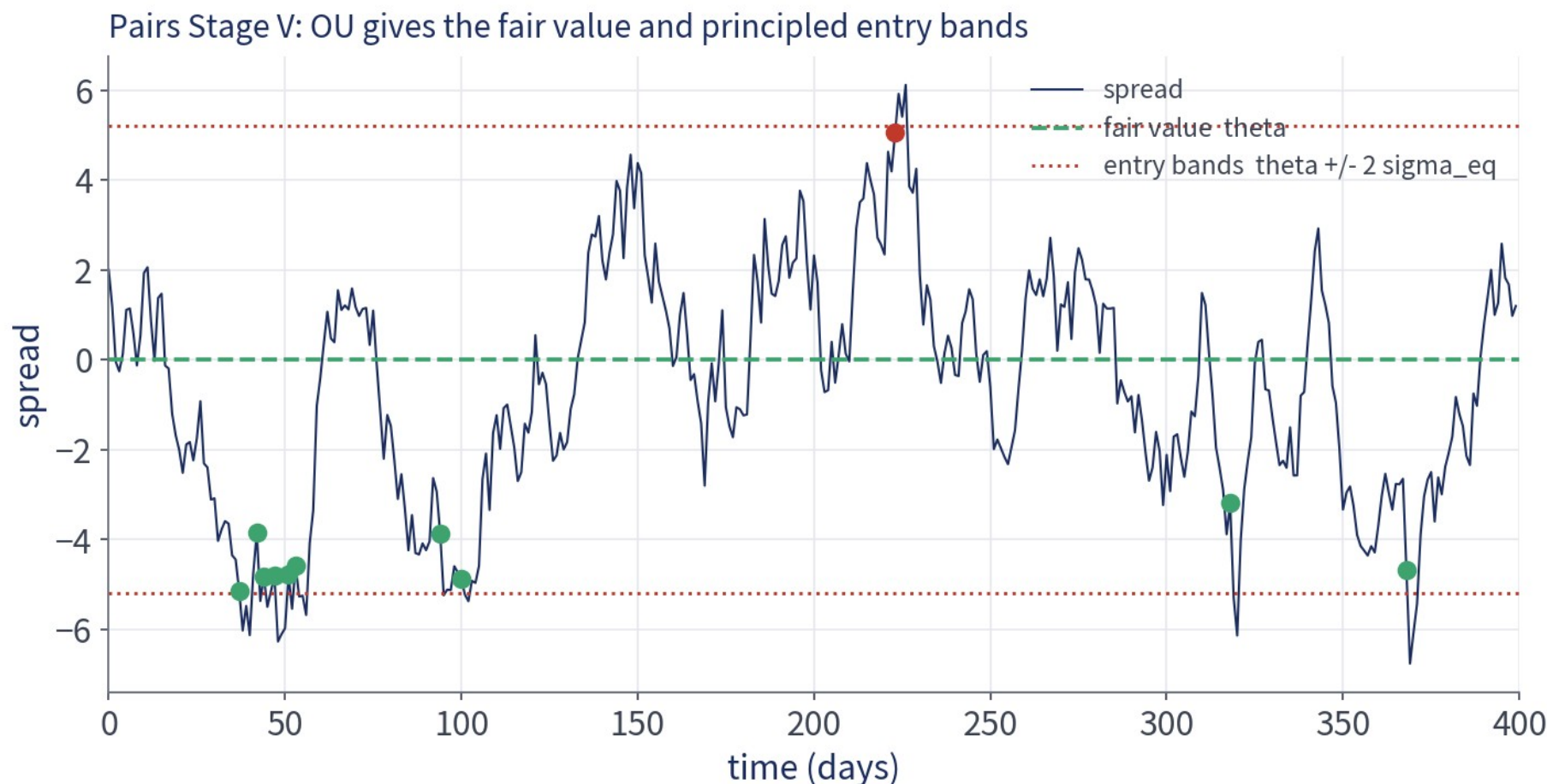
- Enter when the spread is far from theta ...
- ... measured in units of its long-run spread.
- That spread is sigma over the root of twice kappa.

The exit

- Exit as the spread reverts back toward theta.
- The half-life tells you roughly how long that takes.
- Bands and horizon both come from the fit.

Stage V: OU Bands on the Spread

The fitted OU gives a fair value (theta) and entry bands set by the spread's stationary spread. Fade the spread when it crosses a band and exit as it reverts — a model-based mean-reversion trade.



What OU Adds Over Earlier Stages

Principle where there were heuristics

Before Stage V

- Bands were fixed z-scores, chosen by hand.
- No sense of how long a trade should last.
- No explicit estimate of fair value.

With Stage V

- Bands scale with the spread's own statistics.
- The half-life gives a natural holding horizon.
- Theta is an explicit, estimated fair value.

Estimating the OU Parameters

Three numbers from a spread series

Fitting OU means recovering kappa, theta, and sigma from a spread. The cleanest route uses the AR(1) bridge from Part 2: regress the spread on its lag, then translate the persistence into a reversion speed and half-life.

The routes

- AR(1) regression — the discretization bridge.
- Maximum likelihood on the OU transition density.
- They agree; the regression is the simplest.

What you get out

- Theta from the regression's implied mean.
- Kappa from the persistence coefficient.
- Sigma and the half-life follow directly.

Estimation via the AR(1) Bridge

A one-line regression, then translate

$$\text{estimate via AR(1): } \phi \Rightarrow \kappa = -\frac{\ln \phi}{\Delta t}, \quad \theta, \quad t_{1/2} = \frac{\ln 2}{\kappa}$$

The steps

- Regress the spread on its own one-step lag.
- The slope is the AR(1) persistence ϕ .
- Convert ϕ into κ , and thus a half-life.

Why it's convenient

- It is a single ordinary regression.
- It uses the exact OU-to-AR(1) mapping.
- Fast enough to run across many candidate pairs.

The Half-Life as a Trade Filter

Screen pairs by how fast they revert

Too fast

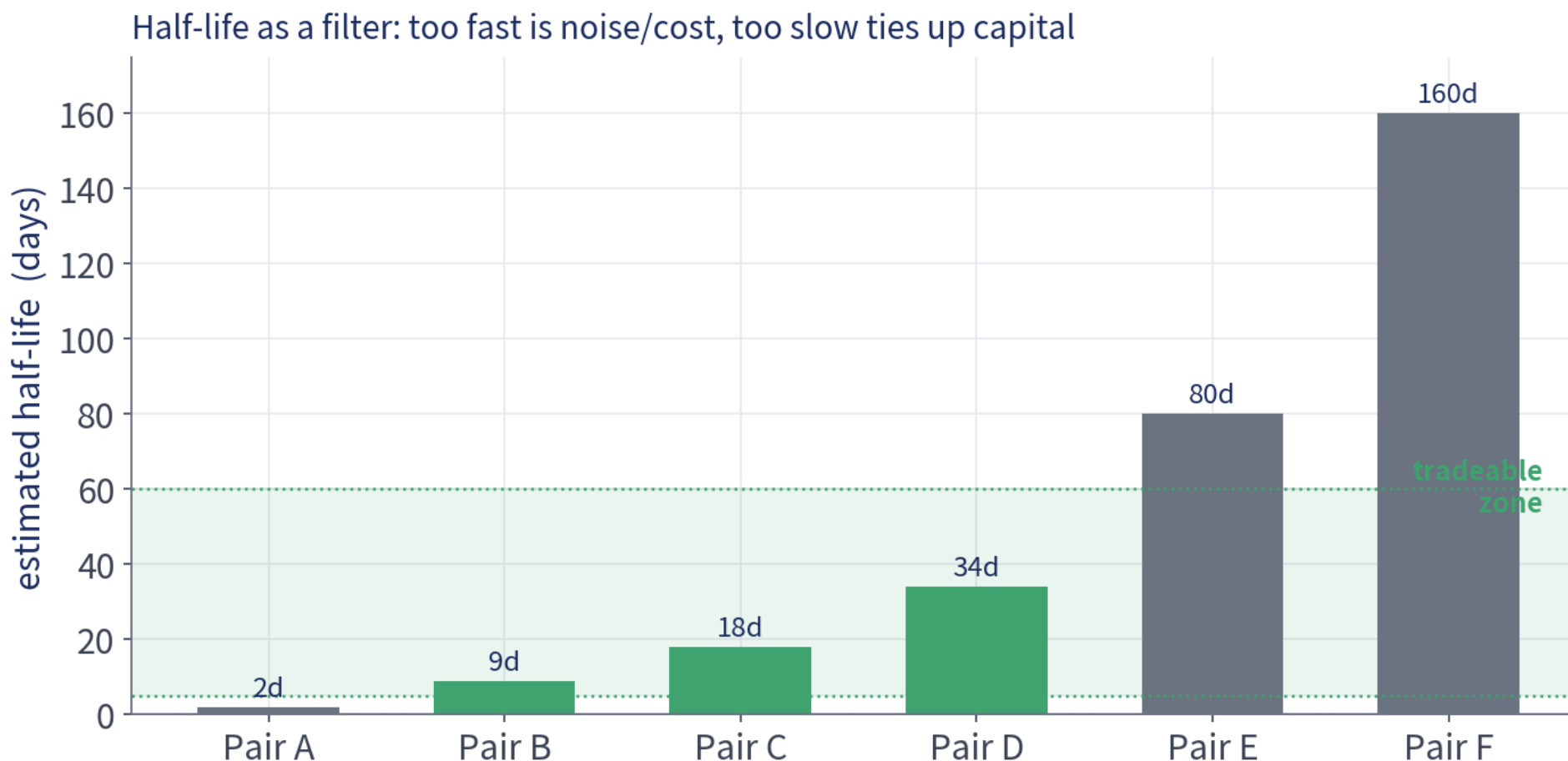
- A very short half-life reverts within noise.
- Trading costs eat any edge.
- Often it is just microstructure, not signal.

Too slow

- A very long half-life ties up capital for ages.
- Reversion may not complete before things change.
- Prefer a sensible, tradeable middle range.

Half-Life as a Filter

Estimated half-lives across candidate pairs. Too short is noise and cost; too long ties up capital. A tradeable zone (shaded) keeps only pairs that revert on a usable horizon.



Estimation and Cautions

Fitting OU is easy; trusting it is the hard part

You can always fit an OU to a spread — the regression never fails. The hard part is trusting the fit. If the spread isn't truly stationary, or its parameters drift, the tidy fair value and bands can be dangerously misleading.

The easy part

- The regression always returns some κ , θ , σ .
- You get a fair value and bands automatically.
- It all looks clean and authoritative.

The hard part

- Is the spread actually stationary right now?
- Are the parameters stable, or drifting?
- A fit on a broken spread is worse than no model.

Non-Stationarity and Regime Breaks

When the spread stops reverting

What goes wrong

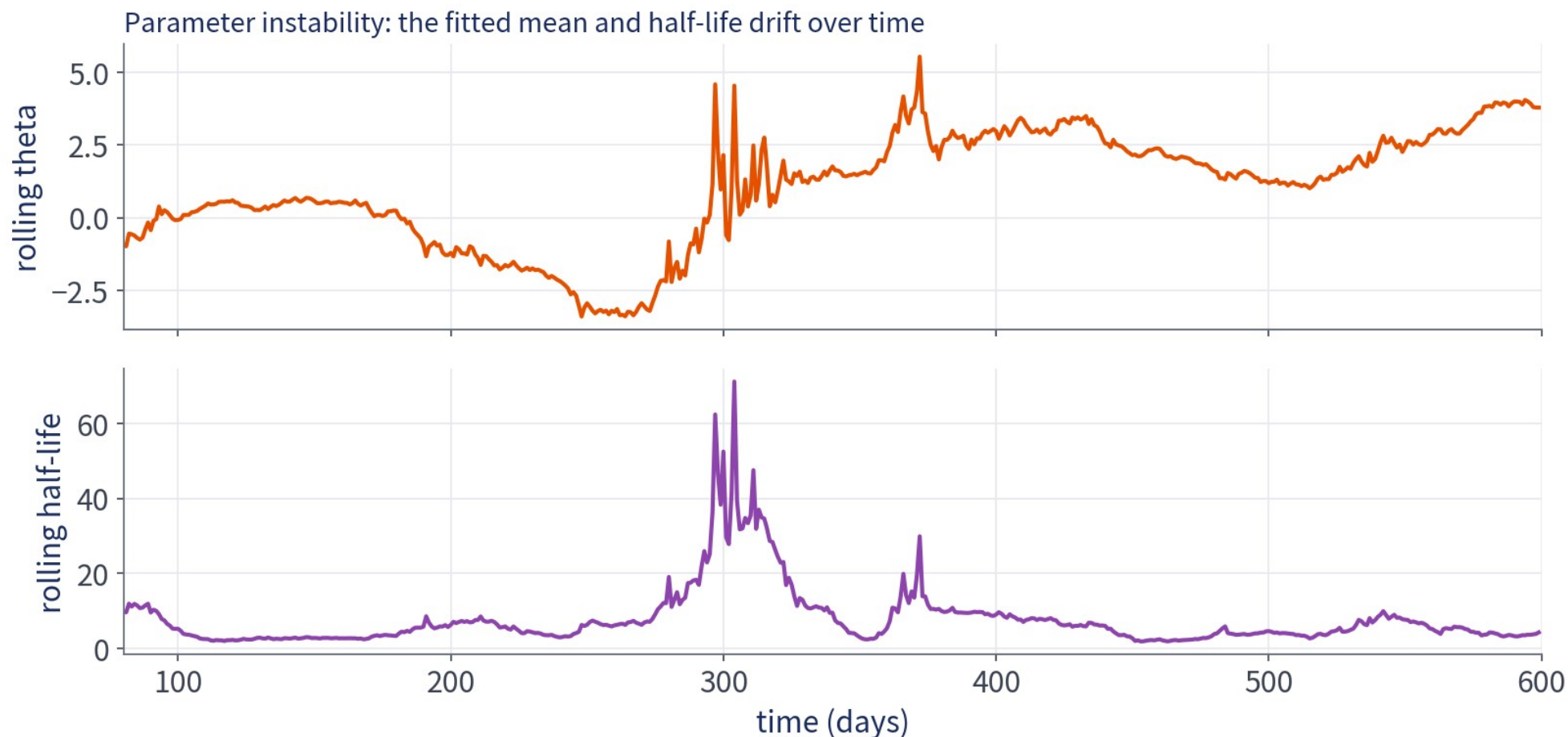
- The cointegration can break (Week 7).
- In a crisis the spread may trend, not revert (Week 10).
- OU then models an equilibrium that no longer exists.

How to guard

- Re-test stationarity, don't assume it persists.
- Overlay the regime filter from Stage IV.
- Stand down when the mean-reversion premise fails.

Parameters Drift Over Time

Re-estimating OU on a rolling window shows the fair value and half-life wandering as conditions change. A single static fit hides this — treat the parameters as moving targets, not constants.



Living With Drifting Parameters

Re-estimate, and stay humble

Practical habits

- Re-estimate the OU parameters on a rolling window.
- Watch the half-life; a lengthening one warns of trouble.
- Size positions down as uncertainty rises.

The mindset

- Treat θ and κ as slowly moving, not fixed.
- Combine OU with the earlier stages, not in isolation.
- The model informs the trade; it does not command it.

The Pairs Journey, Now Five Stages

Volatility bands (W5), a cointegration anchor (W7), a Kalman hedge (W9), a regime filter (W10), and now an OU model (W11) that turns the spread into a fair value, bands, and a half-life.

The Pairs Journey, Now Five Stages

Five weeks, five layers - Stage V gives the spread a principled continuous-time model.



What Stage V contributes

Earlier stages built and protected the spread; Stage V finally MODELS it. The OU process gives a fair value (theta), statistically-grounded entry bands (from the stationary spread), and a half-life that sets the horizon.

From ad-hoc thresholds to a principled, estimable model of mean reversion.

The Continuous-Time Toolkit

Everything Week 11 gave you

Object	Form	Use
Brownian motion	dW	raw continuous randomness
Geometric BM	$dS = \mu S dt + \sigma S dW$	asset prices (positive, lognormal)
Ornstein-Uhlenbeck	$dX = \kappa(\theta - X)dt + \sigma dW$	mean-reverting spreads
Ito's lemma	adds a second-order term	functions of a process

Four objects, one language: continuous-time models built on Brownian motion.

When OU Is the Right Model

Match the model to the series

OU fits when

- The series genuinely reverts to a stable mean.
- The reversion is roughly linear in the gap.
- Shocks are roughly Gaussian and constant in size.

OU is wrong when

- The series trends or has a unit root (use GBM/ARIMA).
- Reversion speed or mean shifts sharply (add regimes).
- Jumps or fat tails dominate (need richer models).

Trade the Spread as an OU Process

Direct Claude Code to build the Stage V trade

GOAL Fit an OU model to a pairs spread, derive bands and a half-life, and back-test the mean-reversion trade.

[INPUT] Prompt · Claude Code

Write and run `pairs_stage5.py` that:

1. builds a cointegrated spread and fits OU via the AR(1) regression (recover κ , θ , σ , and the half-life);
2. sets fair value θ and entry bands at $\theta \pm z \cdot \sigma_{eq}$ ($\sigma_{eq} = \sigma / \sqrt{2 \kappa}$);
3. back-tests: enter beyond a band, exit near θ ;
4. re-estimates OU on a rolling window and plots how θ and the half-life drift.

Report the half-life and whether the pair is tradeable, and flag when parameters look unstable.

How to work with it

- Bands should come from the fit, not a guess.
- Half-life sets the holding horizon.
- Watch rolling θ and κ for drift.
- Combine with Stage IV regimes in crises.

Deliverable: `pairs_stage5.py` + OU fit, model-based bands, a back-test, and a drift check.

Three Questions Before You Finish

If you can answer these, Week 11 has landed

Q1 In Stage V, what plays the role of fair value?

The OU mean theta

Q2 How do you estimate OU from a spread most simply?

An AR(1) regression

Q3 Why is fitting OU easy but trusting it hard?

Stationarity can fail

All three come from the OU spread model, the AR(1) estimator, and the stationarity caution.

Three Pitfalls to Avoid

Where Stage V goes wrong

Common mistakes

- "A clean OU fit means the trade is safe."
- "The fitted θ and κ are constants."
- "Any half-life is tradeable."

The reality

- A fit on a broken spread is worse than none.
- Parameters drift — re-estimate and watch them.
- Screen out half-lives that are too short or too long.

Week 11 in Three Ideas

Pulling the whole week together

1

Continuous time, built on Brownian motion

Parts 1-2: Brownian motion and the increment dW , SDEs as drift plus diffusion, GBM for prices, Ito's lemma, and the OU process — continuous-time mean reversion with a spring toward the mean.

2

OU is a continuous-time AR(1)

Its half-life is log-two over the reversion speed, and sampling it on a grid gives an AR(1). So you estimate it as a simple regression and read off the continuous parameters.

3

Stage V models the spread

The pairs spread becomes an OU process: a fair value, statistically-grounded bands, and a half-life filter. Fitting is easy; trusting it — checking stationarity and drift — is the real work.

Continuous time gave the spread a principled model — and a language for what comes next.

How It All Connects

Eleven weeks, one thread

The through-line

- AR(1) reversion (W2) -> cointegration (W7) -> Kalman (W9).
- Regimes (W10) -> a continuous-time OU model (W11).
- Each week added a layer to the same pairs trade.

The recurring ideas

- Stationarity, mean reversion, and hidden states.
- Estimate discretely, interpret through a model.
- Every tool is judged by whether it survives real data.

Where We Go From Here

Week 12 · Interest-Rate Models and Discretization

1

Interest-rate models are OU's cousins

The classic short-rate models (Vasicek and its relatives) are mean-reverting SDEs — the OU process, applied to interest rates. Week 11 is the foundation for all of them.

2

Discretization becomes central

To simulate and estimate these SDEs, you discretize them. The exact OU discretization and the Euler scheme are the tools we sharpen next.

3

From one spread to whole curves

We move from modeling a single mean-reverting spread to modeling interest rates and the term structure — continuous-time finance in full.

See you in Week 12 — mean-reverting SDEs for interest rates, and how to discretize them.